Exploratory Data Analysis: Executive Summary

DecodeLabs Data Analytics Internship, Project 2

1. Problem Statement

What do 1,200 customer orders (January 2023 to June 2025) tell us about order values, product performance, sales trends and order outcomes?

2. Methodology

I used the cleaned dataset from Project 1 (0 duplicate IDs, 0 incorrect dates, 0 blank cells). In Google Colab with Python (pandas), I calculated descriptive statistics (count, mean, median, quartiles), detected outliers with the IQR method (below Q1 minus 1.5 x IQR or above Q3 plus 1.5 x IQR), checked correlations between Quantity, UnitPrice and TotalPrice, and compared products, months and order statuses.

3. Key Findings

Typical order value is lower than the average suggests. The mean order is 1,053.97 but the median is 823.62, so the data is right-skewed. A few large orders pull the average up, and the median is the better measure of a typical order.

8 high-value outliers are genuine orders, not errors. All sit above about 3,330 and are 5-unit orders of expensive items. I kept them and flagged them as high-value orders worth investigating.

Price drives order value slightly more than quantity. UnitPrice has a correlation of 0.72 with TotalPrice and Quantity has 0.62. Quantity and UnitPrice are unrelated (0.01). Correlation is a clue, not proof of cause.

Chair and Printer earn the most; Phone earns the least. Chair and Printer each bring in about 195.6k, while Phone brings in about 151.7k. Order counts are evenly spread across the 7 products (156 to 181 orders each).

Only about 19% of orders were delivered. Of 1,200 orders: Cancelled 250 (20.8%), Returned 247 (20.6%), Pending 237 (19.8%), Shipped 235 (19.6%), Delivered 231 (19.3%). Cancelled and returned orders together make up about 41% of all orders.

First-half revenue fell each year. January to June revenue was about 286.5k in 2023, 257.1k in 2024 and 231.9k in 2025, a drop of roughly 10% each year. Month-to-month revenue is otherwise irregular, with no steady seasonal pattern.

4. Recommendations

1. Investigate why 41% of orders are cancelled or returned. Reasons are not in this dataset, so collect them.
2. Study the 8 high-value orders and the Chair and Printer lines to see what can be repeated.
3. Look into the decline in first-half revenue before planning next year's targets.

Limitations

Revenue figures use TotalPrice for all orders, including cancelled and returned ones, so real revenue is likely lower. TotalPrice is calculated before coupon discounts. This analysis shows patterns, not causes.

Tools: Google Sheets, Google Colab, Python (pandas)